

Bodies in Space

Interactive Environments

Lecture 4 — Interaction Design for Artists

NABA Milan · 2025–26

The Shift

LECTURES 1–3

The World Lives On Its Own

Autonomous creatures

Emergence from simple rules

Your presence is a disturbance

“What happens without me?”

LECTURE 4

The World Responds To You

Reactive environments

Body as interface

Your presence is the input

“What happens because of me?”

REFERENCE

Daan Roosegaarde — Dune

2006 – ongoing · Rotterdam, NL

Hundreds of light fibers line a public walkway.

They glow and hum as you walk past.

They dim when you leave.

Presence → Light + Sound

One sensor. One response. Hundreds of copies.
That's it. And it feels alive.

[image: Dune installation]

REFERENCE

Random International — Rain Room

2012 · Barbican, London

Rain falls everywhere in a dark room.

Except where you stand.

Cameras track your position.

Water nozzles shut off above you.

Presence → Absence

The space doesn't add something.

It removes something. That's still a response.

[image: Rain Room]

REFERENCE

Chris Milk — The Treachery of Sanctuary

2012 · Creators Project, New York

Three shadow panels. Same input: your body.

Panel 1: your silhouette dissolves into birds

Panel 2: you grow wings

Panel 3: you flap your arms, release birds

Body → Escalating Transformation

The design isn't the technology.
It's the mapping between action and response.

[image: Treachery of Sanctuary]

**What does the space
know about you?**

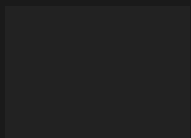
And what does it do with that knowledge?

How the Body Speaks

Four things the space can sense about you

PRESENCE

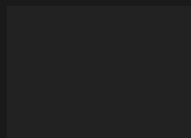
You're near something



ProximitySensor

GAZE

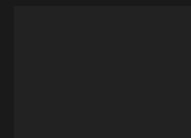
You're looking at something



GazeSensor

CONTACT

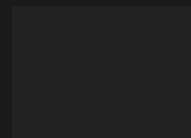
You're touching something



TriggerSensor

INTERACTION

You picked something up



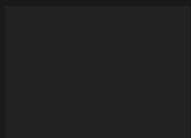
Pickup (new)

How the Space Answers

Four ways the environment can respond

LIGHT

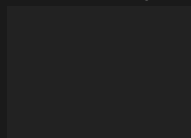
Brightness, color,
shadow, glow



Light + Emission

SURFACE

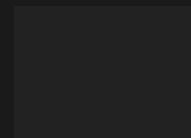
Material color,
texture,
transparency



MaterialColor
MaterialFloat

SOUND

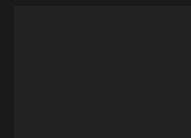
Spatial audio,
ambient tones



SoundPlayer (new)

ATMOSPHERE

Fog, post-
processing,
skybox, mood



AtmosphereControl
(new)

The Design Space

Any input can drive any output — the mapping is the design

	Light	Surface	Sound	Atmosphere
Presence	Pillar glow	?	?	?
Gaze	?	Panel reveal	?	?
Contact	?	Tile color	Tile sound	?
Pickup	?	?	?	Room transform

16 possible mappings. Today we build 4. You choose the rest.

From Wiring to Specifying

A new way to create behaviors

BEFORE: Inspector Wiring

[Pillar]

- └ ProximitySensor
- └ SensorResponse
- └ MaterialEmissionIntensity (on)
- └ MaterialEmissionIntensity (off)
- └ ??? (no light control)

5 components, 6 event wires, can't control light

AFTER: Spec-Driven Controller

[Pillar]

- └ ProximitySensor
- └ GlowOnPresence

2 components. Name = behavior.

Handles emission + light + animation.

You describe what you want.

AI writes the code.

You wire 2 references.

The Spec

Plain language → describe references, config, behavior → AI generates the script

GlowOnPresence

References

- ProximitySensor (on same object)
- Renderer (mesh renderer)
- Light (child point light)

Config

- emissionColor (Color, default warm orange)
- activeIntensity (float, default 3.0)
- idleIntensity (float, default 0.0)
- fadeSpeed (float, default 0.5)

Behavior

- When sensor detects: animate emission + light to active values
- When sensor loses all: animate back to idle values
- Use DOTween for smooth animation

Today's Build: The Responsive Room

1

Presence → Light

Pillars glow when you walk near

ProximitySensor + GlowOnPresence

2

Gaze → Surface

Wall panels reveal when you look at them

GazeSensor + RevealOnGaze

3

Contact → Sound

Floor tiles play tones when you step on them

TriggerSensor + SoundOnContact

4

Pickup → Atmosphere

Crystals transform the entire room

Pickup + AtmosphereControl

One room. Four layers. Each layer is a spec you write → AI generates → you wire.

The mapping is the art.

Same room. Same walls. Same pillars.

But what the space knows about you — and what it does with that —
changes everything.

Dune fills one cell in the matrix and makes it extraordinary.
You don't need sixteen. You need the right one.